

Appendix B: Viral-Vector Tier Selection Decision Table

This appendix is a conceptual governance aid for Section 7. It does not replace nationally recognised biosafety standards, institutional biosafety committee review, a qualified biosafety officer, or applicable law. The recognised containment requirement for the parent agent, vector system, and procedure is always the minimum floor.

Use rule: determine the provisional tier by the highest triggered row. Two or more concurrent escalation factors, or an unresolved critical uncertainty, require independent review and normally one additional tier of conservatism. Institutional review may escalate the result; it may not downgrade it.

B.1 Decision sequence

1. Establish the recognised baseline for the parent agent, vector system, procedure, scale, and host.
2. Evaluate every factor in Table B-1; do not average risks across rows.
3. Select the highest provisional tier triggered by any row.
4. Add an uncertainty adjustment when key characteristics are unknown, disputed, or supported only by weak evidence.
5. Require an independent biosafety and legal review before work begins or changes tier.
6. A Tier 7 boundary result means stop or prohibit the work unless a validated, legally authorised facility and governance framework actually exist. If they do not exist, the work does not proceed.

Viral-vector decision aid

Decision factor	Lower-concern profile	Escalation logic and provisional result within this proposal
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Recognised baseline	Well-characterised vector and procedure governed by an existing national or institutional standard.	Floor: use the recognised requirement. The proposed higher tiers apply only when added characteristics below trigger escalation.
Replication status	Replication-defective system; rescue pathway addressed; controls and inactivation are validated.	Tier 4 Enhanced: lower-concern profile within the report's scope. Tier 5 Remote: conditional replication or plausible helper rescue. Tier 6 Automated: replication-competent or poorly bounded control. Tier 7 boundary: replication combined with efficient human spread, severe outcome, and no reliable countermeasure.
Host range and tropism	Narrow, stable host range with no intended increase in human or mammalian tropism.	Tier 5: expanded but bounded host range. Tier 6: broad, uncertain, or newly human-relevant tropism. Tier 7 boundary: deliberate broadening intended or predicted to support efficient human transmission with severe disease.
	No credible respiratory	Tier 5: aerosol generating procedure or

Decision factor	Lower concern profile	Escalation logic and provisional result within this proposal
Transmission and shedding	transmission; no meaningful shedding expected; exposure opportunities are limited and controlled.	Tier 5: aerosol-generating procedure or unresolved shedding question. Tier 6: credible respiratory spread, pre-symptomatic shedding, or asymptomatic shedding. Tier 7 boundary: efficient respiratory spread combined with severe outcome, persistence, or absent countermeasures.
Payload or resulting phenotype	Reporter, therapeutic, or other payload with no expected increase in virulence, immune escape, or systemic dissemination.	Tier 5: bounded immune modulation or cytotoxic effect. Tier 6: severe virulence, neurotropism, immune escape, or systemic dissemination. Tier 7 boundary: stacked severe traits intended or predicted to create pandemic-scale consequences.
Environmental persistence	Rapidly inactivated under validated conditions; limited environmental persistence and well-defined waste controls.	Tier 5: incomplete inactivation evidence or moderate persistence. Tier 6: prolonged persistence, prolonged shedding, or unresolved waste pathway. Tier 7 boundary: persistence combined with efficient respiratory spread and severe outcome.
Countermeasures	Effective vaccine, antiviral, post-exposure measure, and validated decontamination are available for the relevant hazard.	Tier 5: partial or access-limited countermeasure. Tier 6: absent, unvalidated, or materially uncertain countermeasure. Tier 7 boundary: no effective countermeasure combined with efficient human spread and severe disease.
Procedure, scale, and exposure opportunity	Small-scale work in closed systems with validated engineering controls and limited personnel interaction.	Tier 5: animal work, aerosol-generating manipulation, or scale-up that materially increases exposure opportunities. Tier 6: high-throughput or repeated operations where a single containment lapse could seed wider exposure. Tier 7 boundary: no feasible terrestrial workflow can bound the consequence.
Data quality and uncertainty	Critical characteristics are independently characterised with reproducible evidence.	No downgrade for uncertainty. One material gap normally escalates review. Conflicting evidence or unknown critical traits normally require Tier 6 review or a pause . When risk cannot be bounded and no safe lower-tier path exists, classify as Tier 7 boundary / do not proceed .
Research objective	Objective does not seek increased transmissibility, severity, environmental persistence, immune escape, or broad human host range.	Tier 6 minimum: the stated objective materially increases one major hazard dimension. Tier 7 boundary / prohibition: the objective explicitly seeks a highly transmissible, severe, persistent, or countermeasure-resistant human pathogen phenotype.

B.2 Decision record

Every tier-selection decision should preserve: vector or construct identifier; recognised baseline; highest triggered factor; evidence status; uncertainty adjustment; provisional tier; independent reviewers; and final disposition.

Interpretation note: Tier 7 remains a conceptual boundary condition in this draft. A Tier 7 outcome is therefore primarily a decision not to proceed under existing conditions, not permission to improvise an unavailable facility.